

Axioms of Probability
1. Non-negativity: For any event A , $P(A) \geq 0$.
2. Normalization: $P(S) = 1$, where S is the sample space.
3. Additivity: For any two mutually exclusive events A and B , $P(A \cup B) = P(A) + P(B)$. This can be extended to $P(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n P(A_i)$ for mutually exclusive events $A_1, A_2, \dots, A_n$.

Basic Properties of Probability:
$\forall A \in S, P(A) < 1$
$P(A^c) = 1 - P(A)$
$P(\emptyset) = 0$
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
$A \subseteq B \implies P(A) \leq P(B)$

Counting:
Permutation — An arrangement of objects in a specific order.
$P(n, r) = \frac{n!}{(n-r)!}$
Combination — A selection of objects without regard to the order. $C(n, r) = \frac{n!}{r!(n-r)!}$

Binomial Coefficient — The number of ways to choose k successes in n independent Bernoulli trials. $\binom{n}{k} = C(n, k) = \frac{n!}{k!(n-k)!}$
Number of Subsets — The number of subsets of a set with n elements is 2^n .

Conditional Probability:
$P(A B) = \frac{P(A \cap B)}{P(B)}$
Multiplication Rule: — $P(A \cap B) = P(A B)P(B) = P(B A)P(A)$
$P(\bigcap_{i=1}^n A_i) = P(A_1)P(A_2 A_1)P(A_3 A_1 \cap A_2) \dots P(A_n \bigcap_{i=1}^{n-1} A_i)$
Law of Total Probability: — If $B_1, B_2, \dots, B_n$ are mutually exclusive and exhaustive events, then for any event A : $P(A) = \sum_{i=1}^n P(A B_i)P(B_i)$
Bayes' Theorem: —
$P(B_j A) = \frac{P(A B_j)P(B_j)}{P(A)} = \frac{P(A B_j)P(B_j)}{\sum_{i=1}^n P(A B_i)P(B_i)}$

Independence:
Two events A and B are independent if and only if $P(A B) = P(A)$ and $P(A \cap B) = P(A)P(B)$.
If A and B are independent, then B and A , A and B^c , A^c and B , A^c and B^c are also independent.

Random Variables A RV is a function that assigns a real number to each outcome in the sample space of a random experiment.
Discrete RV — A random variable that can take on a countable number of distinct values.
Continuous RV — A random variable that can take on any value within a given range or interval.
$X : S \rightarrow \mathbb{R}$ Where S is the event space and X is the random variable.
PMF — $f(x) = P(X = x)$ (discrete)
$P(a \leq X \leq b) = \int_a^b f(x)dx$ (continuous)

CDF — $F(x) = P(X \leq x) = \sum_{y \leq x} P(X = y)$ (discrete)
$F(x) = P(X \leq x) = \int_{-\infty}^x f(t)dt$ (continuous)
$P(a \leq X \leq b) = F(b) - F(a)$
Bernoulli Random Variable — A discrete RV that takes the value 1 with probability p and the value 0 with probability $1 - p$.
Expected Value — $E(X) = \mu_X = \sum_x xP(X = x)$ or $= \int_{-\infty}^{\infty} xf(x)dx$
$E(h(X)) = \sum_x h(x)P(X = x)$ or $E(h(X)) = \int_{-\infty}^{\infty} h(x)f(x)dx$
$E(X + Y) = E(X) + E(Y)$
$E(X^2) = \sum_x x^2p(x)$ or $= \int x^2f(x)dx$
Variance — $\sigma_X^2 = E[(X - \mu_X)^2] = E(X^2) - (E(X))^2$ or $\sigma_X^2 = \int (x - \mu_X)^2 f(x)dx$
Standard Deviation — $\sigma_X = \sqrt{Var(X)}$

Binomial Distribution — Models the number of successes in n independent Bernoulli trials, each with success probability p . $X \sim Bin(n, p)$
PMF: — $P(X = k) = b(x; n, p) = \binom{n}{k}p^k(1 - p)^{n-k}$ for $k = 0, 1, \dots, n$ else 0
CDF: — $B(x; n, p) = P(X \leq x) = \sum_{y=0}^{\lfloor x \rfloor} b(y; n, p)$
Mean: — $E(X) = np$
Variance: — $Var(X) = np(1 - p)$

Negative Binomial Distribution — Models the number of trials needed to achieve r successes in a sequence of independent Bernoulli trials, each with success probability p . $X \sim NB(r, p)$
PMF: — $P(X = k) = \binom{k-1}{r-1}p^r(1 - p)^{k-r}$ for $k = r, r + 1, \dots$ else 0
CDF: — $NB(x; r, p) = P(X \leq x) = \sum_{y=r}^{\lfloor x \rfloor} \binom{y-1}{r-1}p^r(1 - p)^{y-r}$
Mean: — $E(X) = \frac{r}{p}$
Variance: — $Var(X) = \frac{r(1-p)}{p^2}$

Geometric Distribution — A special case of the negative binomial distribution where $r = 1$. It models the number of trials needed to get the first success. $X \sim Geo(p)$
PMF: — $P(X = k) = (1 - p)^{k-1}p$ for $k = 1, 2, \dots$ else 0
CDF: — $G(x; p) = P(X \leq x) = 1 - (1 - p)^{\lfloor x \rfloor}$
Mean: — $E(X) = \frac{1}{p}$
Variance: — $Var(X) = \frac{1-p}{p^2}$

Hypergeometric Distribution — Models the number of successes in n draws from a finite population of size N containing K successes, without replacement. $X \sim Hyp(N, K, n)$
PMF: — $h(x; n, N, K) = P(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$ for $\max(0, n - (N - K)) \leq k \leq \min(n, K)$ else 0
CDF: — $H(x; n, N, K) = P(X \leq x) = \sum_{y=\max(0, n-(N-K))}^{\lfloor x \rfloor} \frac{\binom{K}{y}\binom{N-K}{n-y}}{\binom{N}{n}}$
Mean: — $E(X) = n\frac{K}{N}$
Variance: — $Var(X) = n\frac{K}{N}\frac{N-K}{N}\frac{N-n}{N-1}$

Poisson Distribution — Models the number of events occurring in a fixed interval of time or space, given the average rate λ of occurrence.
Used to model the number of times a rare event occurs in a large population.
$X \sim Pois(\lambda)$
PMF: — $P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$ for $k = 0, 1, 2, \dots$ else 0
CDF: — $P(X \leq x) = \sum_{y=0}^{\lfloor x \rfloor} \frac{\lambda^y e^{-\lambda}}{y!}$
Mean: — $E(X) = \lambda$
Variance: — $Var(X) = \lambda$

Normal — $X \sim \mathcal{N}(\mu, \sigma^2)$
PDF: — $f(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$ for $-\infty < x < \infty$ else 0
Standard Normal — $Z \sim \mathcal{N}(0, 1)$ $Z = \frac{X-\mu}{\sigma}$
$P(a \leq X \leq b) = P(\frac{a-\mu}{\sigma} \leq Z \leq \frac{b-\mu}{\sigma}) = \Phi(\frac{b-\mu}{\sigma}) - \Phi(\frac{a-\mu}{\sigma})$
z_α is the z-score such that $P(Z \geq z_\alpha) = \alpha$ or the $(1 - \alpha) \cdot 100$ th percentile

Exponential — $X \sim Exp(\lambda)$
PDF: — $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$ else 0
CDF: — $F(x) = 1 - e^{-\lambda x}$ for $x \geq 0$ else 0
Mean: — $E(X) = \frac{1}{\lambda}$
Variance: — $Var(X) = \frac{1}{\lambda^2}$

Uniform — $X \sim U(a, b)$
PDF: — $f(x) = \frac{1}{b-a}$ for $a \leq x \leq b$ else 0
CDF: — $F(x) = \frac{x-a}{b-a}$ for $a \leq x \leq b$, 0 for $x < a$, 1 for $x > b$
Survival Function: — $S(x) = 1 - F(x) = \frac{b-x}{b-a}$ for $a \leq x \leq b$, 1 for $x < a$, 0 for $x > b$
Mean: — $E(X) = \frac{a+b}{2}$
Variance: — $\sigma^2 = \frac{(b-a)^2}{12}$

Jointly Distributed Random Variables:
 $P(x, y) = P(X = x, Y = y)$
Trick: $P(X = Y) + P(Y > X) + P(Y < X) = 1$
Marginal PMF — $P_X(x) = \sum_y P(x, y)$ and $P_Y(y) = \sum_x P(x, y)$
 X and Y are independent $\iff P(x, y) = P_X(x)P_Y(y)$
Conditional PMF — $P_{Y|X}(y|x) = \frac{P(x, y)}{P_X(x)}$
Expected Value — $E(h(X, Y)) = \sum_x \sum_y h(x, y)P(x, y)$
 $E(h(X, Y)) = \int \int h(x, y)f(x, y)dxdy$
Covariance — $Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E(XY) - E(X)E(Y)$
 $Cov(X, Y) = 0$ if X and Y are independent.
Correlation Coefficient — $\rho_{XY} = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$
 $-1 \leq \rho_{XY} \leq 1$
Properties $Cov(X, X) = Var(X)$ $Cov(X, Y) = Cov(Y, X)$
 $Cov(aX, bY) = abCov(X, Y)$ $Cov(X + Y, Z) = Cov(X, Z) + Cov(Y, Z)$ $\exists a, b \in \mathbb{R} \ni Y = aX + b$

Random Samples and Statistics:
A random sample of size n is a set of n independent and identically distributed (i.i.d.) random variables $X_1, X_2, \dots, X_n$ drawn from a population with a common distribution.
Sample Mean — $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$
Sample Variance — $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$
 $E(\bar{X}) = \mu$
 $Var(\bar{X}) = \frac{\sigma^2}{n}$
When sampling from a normal population, $\bar{X} \sim \mathcal{N}(\mu, \frac{\sigma^2}{n})$

Linear Combinations — $Y = a_1X_1 + a_2X_2 + \dots + a_nX_n$
 $E(Y) = a_1E(X_1) + a_2E(X_2) + \dots + a_nE(X_n)$
 $Var(Y) = a_1^2Var(X_1) + a_2^2Var(X_2) + \dots + a_n^2Var(X_n)$ if X_i are independent.
If $X_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$, then $Y \sim \mathcal{N}(a_1\mu_1 + a_2\mu_2 + \dots + a_n\mu_n, a_1^2\sigma_1^2 + a_2^2\sigma_2^2 + \dots + a_n^2\sigma_n^2)$

Central Limit Theorem:
 $n > 30 \implies \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$
Point Estimation:
Sample Mean: $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$
Sample Proportion: $\hat{p} = \frac{M}{n}$, $M = \#$ of successes
Sample Variance: $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$

Quantitative (μ):
Point Estimator: $\hat{\theta} = \bar{X}$
Bias: $E(\hat{\theta}) - \theta$ Bias of 0 means unbiased estimator
Standard Error: $\sigma_{\hat{X}} = \frac{\sigma}{\sqrt{n}}$
Est Standard Error: $\frac{S}{\sqrt{n}}$
 $Var(\hat{\theta}) = \frac{\theta(1-\theta)}{n}$
 $P(|\bar{X} - \mu| \leq c) = P(-c \leq \bar{X} - \mu \leq c) = P\left(-\frac{c}{\sigma/\sqrt{n}} \leq Z \leq \frac{c}{\sigma/\sqrt{n}}\right)$
 $= \Phi_Z\left(\frac{c}{\sigma/\sqrt{n}}\right) - \Phi_Z\left(-\frac{c}{\sigma/\sqrt{n}}\right) = 2\Phi_Z\left(\frac{c}{\sigma/\sqrt{n}}\right) - 1$
Qualitative (p):
Point Estimator: $\hat{p} = \frac{M}{n}$
Bias: $E(\hat{p}) - p$ Bias of 0 means unbiased estimator
Standard Error: $\sqrt{\frac{p(1-p)}{n}}$
Est Standard Error: $\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

Confidence Intervals:
Confidence Level: $100(1 - \alpha)\%$
 $CI_\mu = (\bar{X} \pm z_{\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}})$, $z_{\frac{\alpha}{2}}$ is the z-score such that $\Phi_Z(z_{\frac{\alpha}{2}}) = 1 - \frac{\alpha}{2}$
 $99\% \rightarrow \alpha = 0.01 \rightarrow z_{\frac{\alpha}{2}} = 2.576$
 $95\% \rightarrow \alpha = 0.05 \rightarrow z_{\frac{\alpha}{2}} = 1.96$
 $90\% \rightarrow \alpha = 0.10 \rightarrow z_{\frac{\alpha}{2}} = 1.645$
Higher confidence levels lead to wider confidence intervals. Larger sample sizes lead to narrower confidence intervals. Larger standard deviations lead to wider confidence intervals.
Confidence Intervals for Large Samples:
If $n \geq 40$ then we can replace σ with S in the confidence interval formula in cases where σ is unknown.
 $Z = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim \mathcal{N}(0, 1)$ approximately by CLT.
 $CI_\mu = (\bar{X} \pm z_{\frac{\alpha}{2}} \cdot \frac{S}{\sqrt{n}})$
 $CI_p = \left(\hat{p} \pm z_{\frac{\alpha}{2}} \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\right)$

Hypothesis Testing:
Statistical Hypothesis — A claim about a population parameter.
Test Statistic — A statistic calculated from sample data that is used to make decisions about the hypothesis.
Null Hypothesis (H_0) — The hypothesis that there is no effect or difference, and it is the hypothesis that is tested.
Alternative Hypothesis (H_a) — The hypothesis that there is an effect or difference.
Conclusion — Reject H_0 or Fail to Reject H_0 .
Type I Error — Rejecting the null hypothesis when it is true.
Type II Error — Failing to reject the null hypothesis when it is false.
Significance Level (α) — The probability of making a Type I error. Common choices are 0.05, 0.01, and 0.10.
Rejection Region — The set of values for the test statistic that leads to the rejection of the null hypothesis. $p\text{-value} \leq \alpha \implies \text{Reject } H_0$

P-Values
The probability of obtaining a test statistic as extreme as the one observed, assuming the null hypothesis is true.
Left-Tailed Test ($H_a : \mu < \mu_0$):
 $p\text{-value} = P(Z \leq z_{obs}) = \Phi_Z(z_{obs})$
Right-Tailed Test ($H_a : \mu > \mu_0$):
 $p\text{-value} = P(Z \geq z_{obs}) = 1 - \Phi_Z(z_{obs})$
Two-Tailed Test ($H_a : \mu \neq \mu_0$):
 $p\text{-value} = 2P(Z \geq |z_{obs}|) = 2(1 - \Phi_Z(|z_{obs}|))$

Tests for a Population Mean:
Case 1: Sampling from a Normal population with known σ and unknown μ . Suppose we want to test $H_0 : \mu = \mu_0$. In this case our test statistic is $Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$. $\bar{X}$ is the sample mean and μ_0 is the hypothesized population mean under H_0 .
Case 2: If σ is unknown and $n \geq 40$, we can use the same test statistic as in Case 1, but replace σ with S (the sample standard deviation). $Z = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$ approximately follows a standard normal distribution by the CLT.

Tests for a Population Proportion:
We want to test $H_0 : p = p_0$. Our test statistic is $Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$. $\hat{p}$ is the sample proportion and p_0 is the hypothesized population proportion under H_0 .